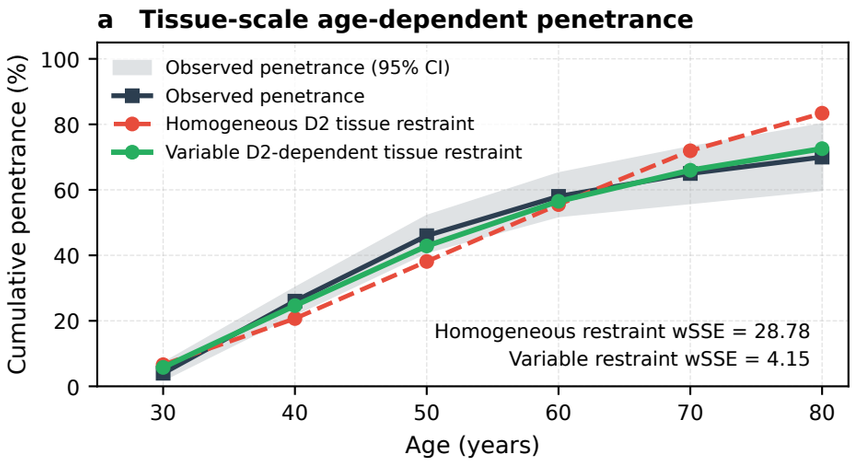

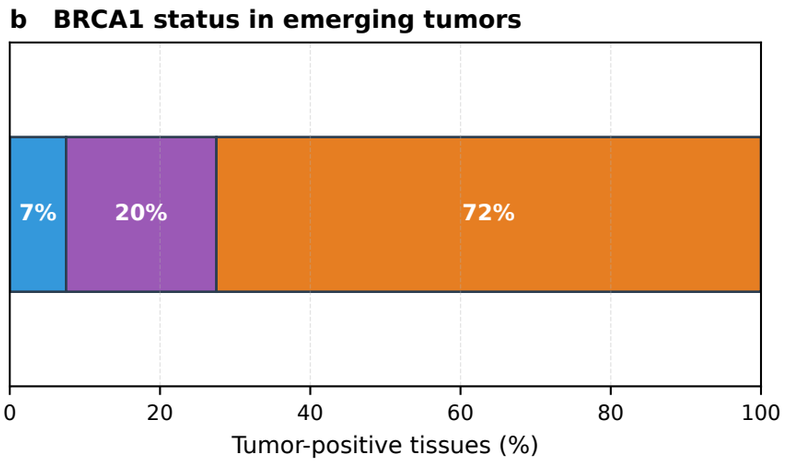

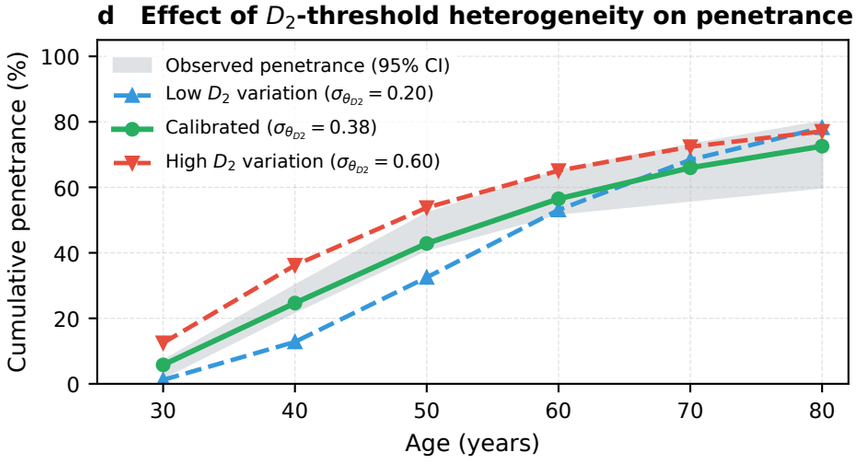

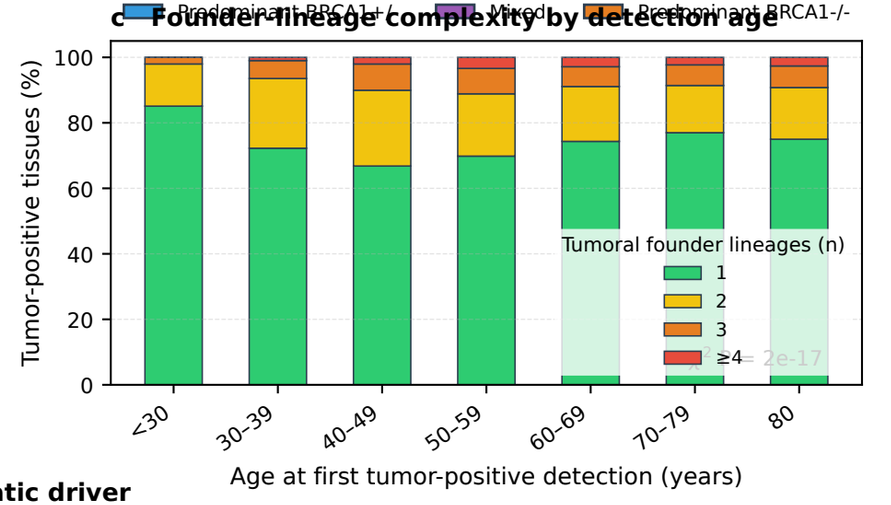

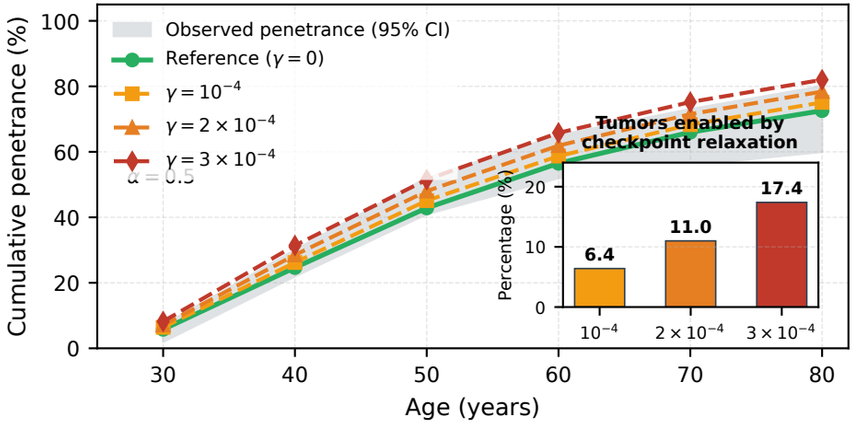

Tissue-Scale Simulation Summary (10^7 cells):

- Clinical Penetrance agreement: wSSE = 4.15
- BRCA1-loss-majority tumors (>50% null): 92.5%
- Subclonal BRCA1+/- retention (≥25%): 21.6%
- Multi-founder tumors (≥2 lineages): 28.1%
- Tissue-restraint variation governs early vs late timing
- Robust to alternative somatic mutation ordering